

Sparkybot: An Embodied AI Agent-Powered Robot with Customizable Characters and Interaction Behavior for Children

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Figure 1: Three platform prototypes: Sparkybot with the interactive accessory of (a) Object Adapter, (b) Character Creator, and (c) Environment Interactor. Two demonstration scenarios: (d) Customizable Character and (e) Interaction Behavior with Sparkybot.

ABSTRACT

Sparkybot is a novel robotic platform designed to enrich children-robot interaction by leveraging embodied agent technology. The platform allows children to customize the interaction characters and behaviors of the robot, providing a more personalized and engaging experience. Two scenarios are demonstrated to showcase how children can customize and interact with the robot. Our goal is to provide a platform that enables children to interact with robots in a more creative and engaging way.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computer systems organization** → **Robotic components**.

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KEYWORDS

robotic platform, children-robot interaction, embodied agent, contextual behaviors

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1 INTRODUCTION

Over the past few years, robots have increasingly played a crucial role in children's development, serving as educational [2], entertaining [1], and companionable entities during interactions with children [7]. However, most of these robots can only perform preset actions, lacking interactivity and personalized experiences. This limitation often leads to children quickly becoming bored and dissatisfied. Some studies have shown that children lose interest in their robots once the novelty effect wears off [9].

Recently, the development of embodied agent technology has brought about a revolutionary change in addressing this issue with these robots. Embodied agents based on Large Language Models

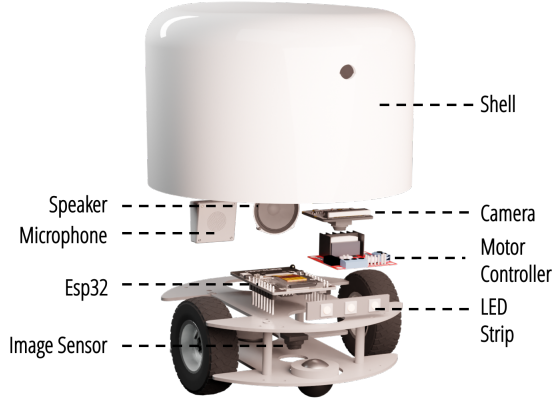


Figure 2: The robot hardware design

(LLMs) have the potential to learn to represent the world through projection into language space [5]. This makes it possible for the robot to automatically plan feedback in real time from the user’s vague commands during the interaction. Using this technology, robots can extract affordances and constraints during conversations with children and give them personalized, non-preset interactions. However, how to design these personalized, non-preset interactions to enhance children’s experience with robots remains fairly unexplored.

In this work, we propose an idea that allows children to customize the interaction characters and behaviors of robots using embodied agents. To this end, we developed Sparkybot, a mobile robot platform that includes a robot, three types of interactive accessory, and an embodied agent. Its embodied agent can interpret the character and behavior it should embody in the interaction from children’s vague commands, autonomously plan the robot’s interaction, and ultimately execute instructions. To demonstrate the Sparkybot platform in interacting with children, we showcase two interaction scenarios: customizable characters for children and customizable interaction behavior with the robot.

2 DESIGN AND IMPLEMENTATION

Our design goal is to allow children to customize the interaction characters and behaviors of the robot, thereby providing a more personalized and engaging experience for children interacting with robots.

Robot Hardware Design. Figure 2 illustrates the hardware design of Sparkybot. The robot is designed as a cylinder, with a platform on top where children can place our interactive accessories or daily objects. The robot captures real-time video streams and dialogue information from children using its built-in camera and microphone. It provides dialogue feedback to children through its speaker and visual feedback via its LED strip. The robot navigates itself on printed sheets of paper “augmented” with a dot pattern[4]. The paper size can be customized according to the activity requirements.



Figure 3: The interactive accessory design

Interactive Accessory Design. Figure 3 explains the design of Sparkybot’s interactive accessories. We developed three types of accessories to support the robot:

- **Character Creator:** This accessory enables children to create a hand-drawn character and interaction behavior for the robot. Children can draw on the accessory, and through dialogue, they can imbue the robot with the personality and interaction behaviors of their created characters.
- **Object Adapter:** This accessory allows children to place daily objects on the robot and have the robot embody the character of these objects for personalized interactions. Children can place their daily items (such as toys, plants, stationery, etc.) on the accessory, and through dialogue, instruct the robot on how they should plan the interaction behavior if they “are” these objects.
- **Environment Interactor:** This accessory provides the robot with the ability to interact with the environment. Children can attach pointers or other objects to the robot, and through dialogue, set up complex interactions such as choice games that involve environment interaction.

Embodied Agent Design. Figure 4 explains the design of Sparkybot’s embodied agent, which helps the robot respond to user commands and perform interaction behaviors. The agent first translates multimodal data obtained by the robot from sensors (e.g., camera, microphone) into text information through a translator module. Subsequently, through a Prompt Structuring module, the agent can understand and analyze the intent contained in children’s commands, and formulate corresponding formatted interaction requirements (e.g., set Characters and interaction behaviors) through parallel reasoning. These formatted pieces of information will be processed by the large language model, assisting the robot in planning and executing the interaction behaviors (e.g., voice feedback and movement). Ultimately, the robot’s behavior will align with the children’s intentions and desires.

3 DEMONSTRATION

Customizable Character with Sparkybot. Inspired by previous research on children’s storytelling with robots [6, 10], we designed a customizable character scenario for Sparkybot. In the demonstration, we presented a scenario where children use an “object adapter” accessory to allow the robot to take on the characters of a turtle and a rabbit placed on it, as shown in figure 1 (d). They can ask the robot to take on the characters through simple conversations, and the embodied agent will then generate voice responses and

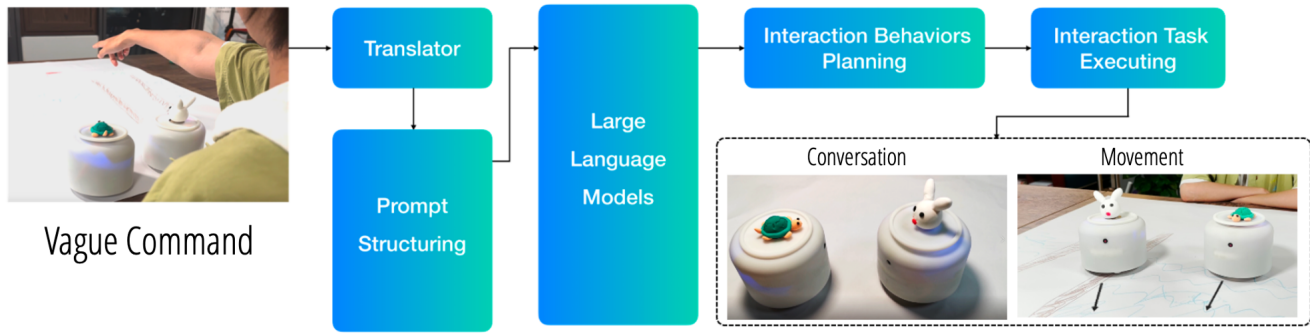


Figure 4: Our embodied agent that helps Sparkybot analyzes children’s vague commands, customizes Characters and interaction behaviors, and performs the corresponding interactions

interactive behaviors that match its character. With this design, we hope that in the future, children will be able to use the robot to actually interact and talk to any object or cartoon character in their imagination.

Customizable Interaction Behavior with Sparkybot. Inspired by previous research on children’s tabletop games with robots [3, 8], we also designed a customizable interaction behaviors scenario for Sparkybot. In the demonstration, we presented a scenario where, after using the “object adapter” to assign characters to the robot, children can place the robot in a preset workspace and tell the robot some game rules and behaviors they imagine through dialogue. For example, as shown in figure 1 (e), the robot assigned the characters of a turtle and a hare can re-enact the fable of The Hare and The Tortoise in the preset workspace according to the children’s design. We expect that in the future, children will be able to spontaneously create and design any interesting interaction with robots they imagine through Sparkybot.

4 DISCUSSION AND FUTURE WORK

This work introduces the idea of a robotic platform, Sparkybot, which allows children to customize the interaction characters and behaviors of the robot using embodied agents. We demonstrate two scenarios to showcase the possibilities that the Sparkybot platform brings to children-robot interaction: (1) Customizable Characters with Sparkybot, and (2) Customizable Interaction Behaviors with Sparkybot. However, while our current agent can adapt to different characters’ dialogues and basic robot movement planning, deconstructing the interaction behaviors designed by children in complex scenarios and executing them in robot movement and positioning remains a challenge. This requires us to further develop and analyze the technology of our agent. In our future work, we will further optimize its architecture and analyze the interaction effect of this platform in complex scenarios through user experiments.

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